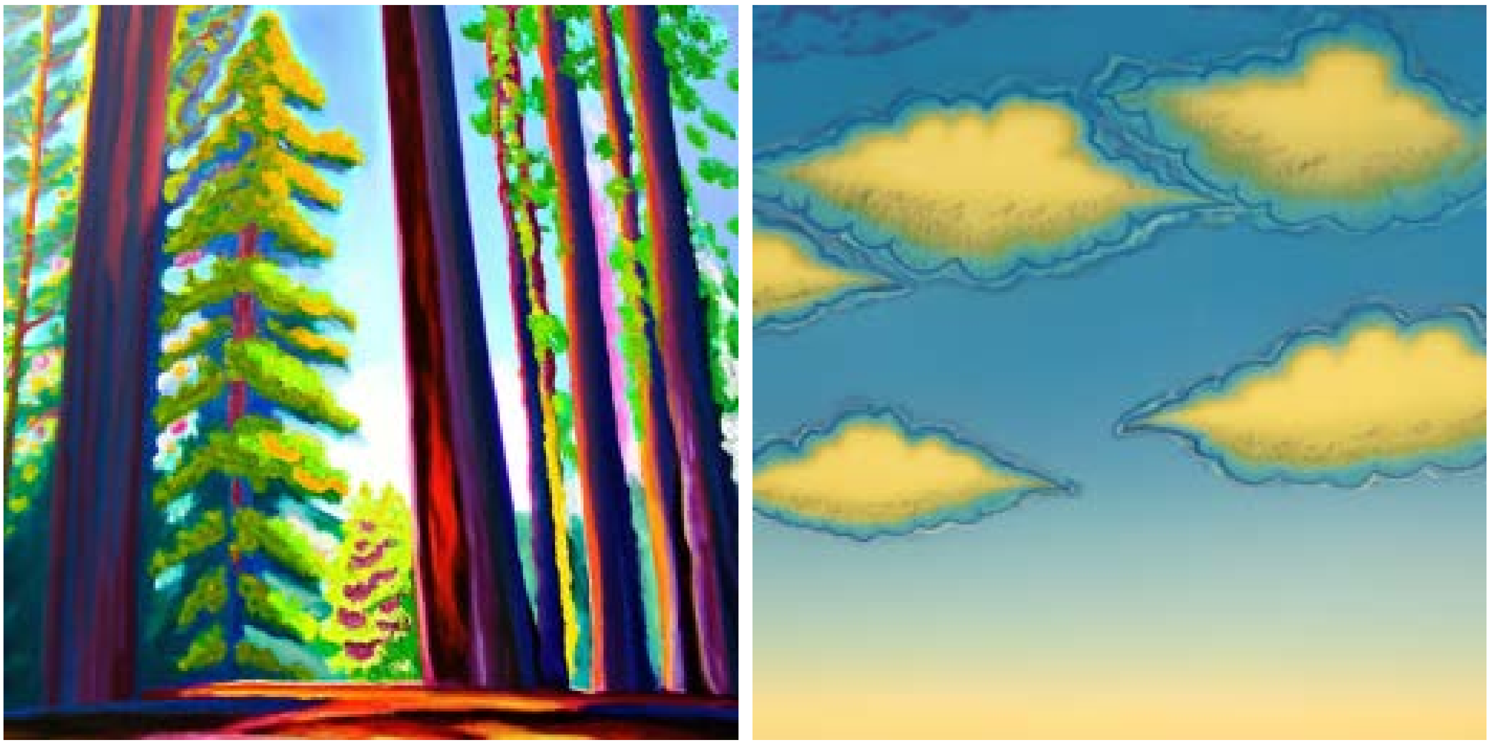
PROJECT FIVE

Part 0: Setup

Below are 3 images captioned with the prompts used to generate them:



"an oil painting of redwood trees"; "a lithograph of clouds"

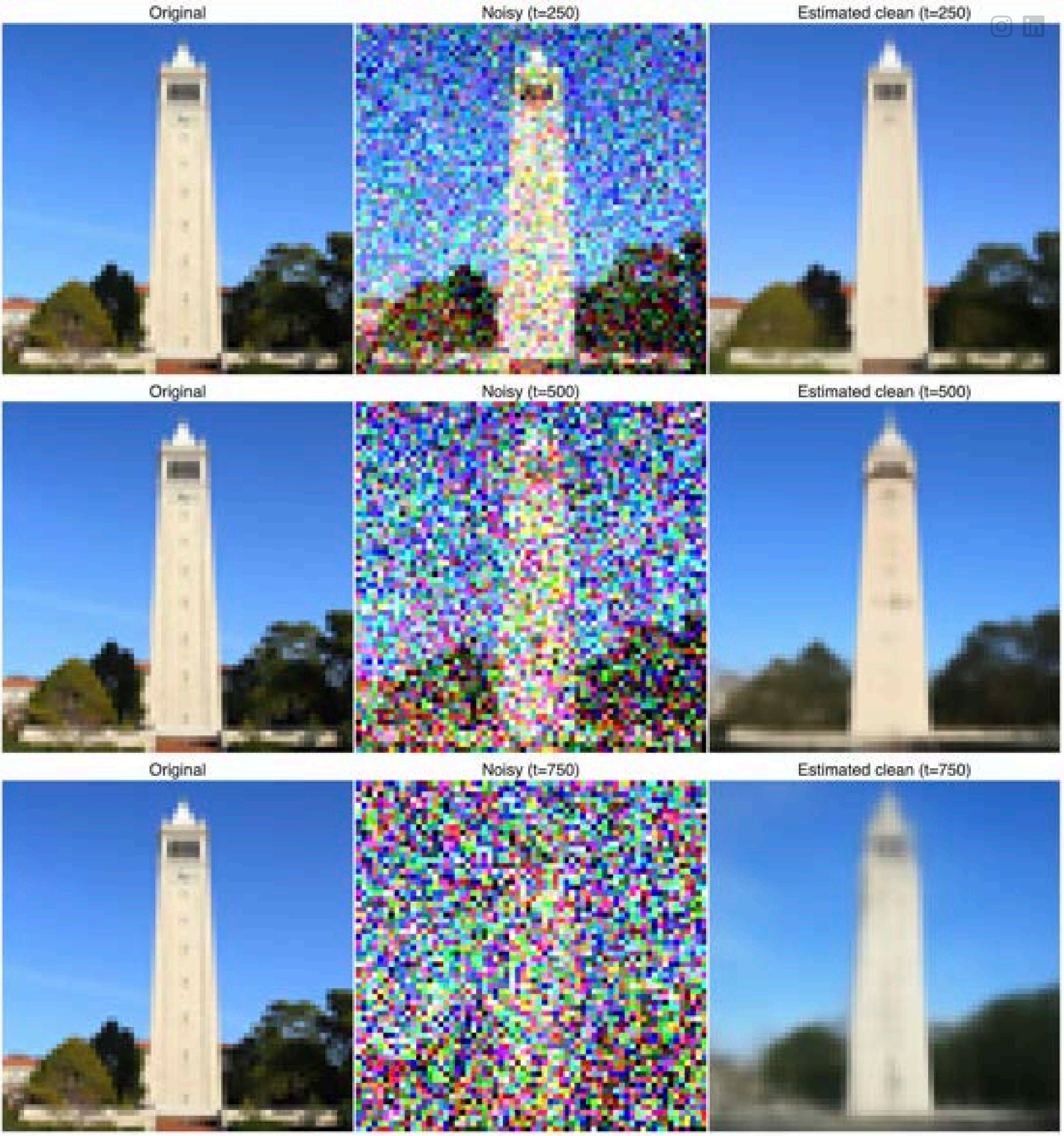
"a photo of the night sky" at 20, 80, and 200 inference steps

Part 1.1, 1.2, & 1.3: Implementing the Forward Process & Denoising

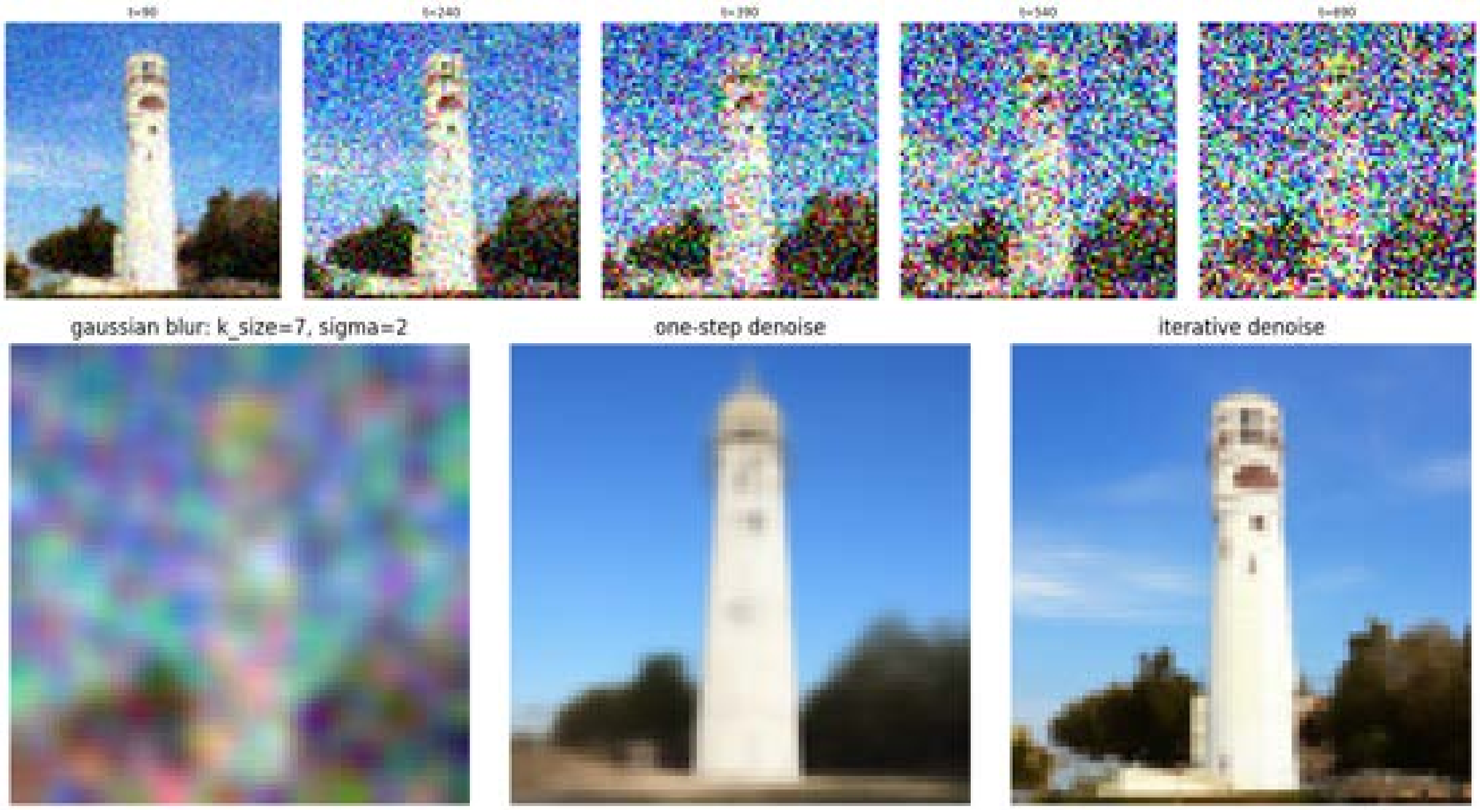
The campanile at noise levels [250, 500, 750]

Gaussian-denoised at levels [250, 500, 750]

Below is the original image, noisy image, and one-step denoised estimate:


Original
Noisy (t=250)
Estimated clean (t=250)
Original
Noisy (t=500)
Estimated clean (t=500)
Original
Noisy (t=750)
Estimated clean (t=750)

Part 1.4: Iterative Denoising



Part 1.5: Diffusion Model Sampling

Below are 5 sampled images, generated from scratch with random noise:

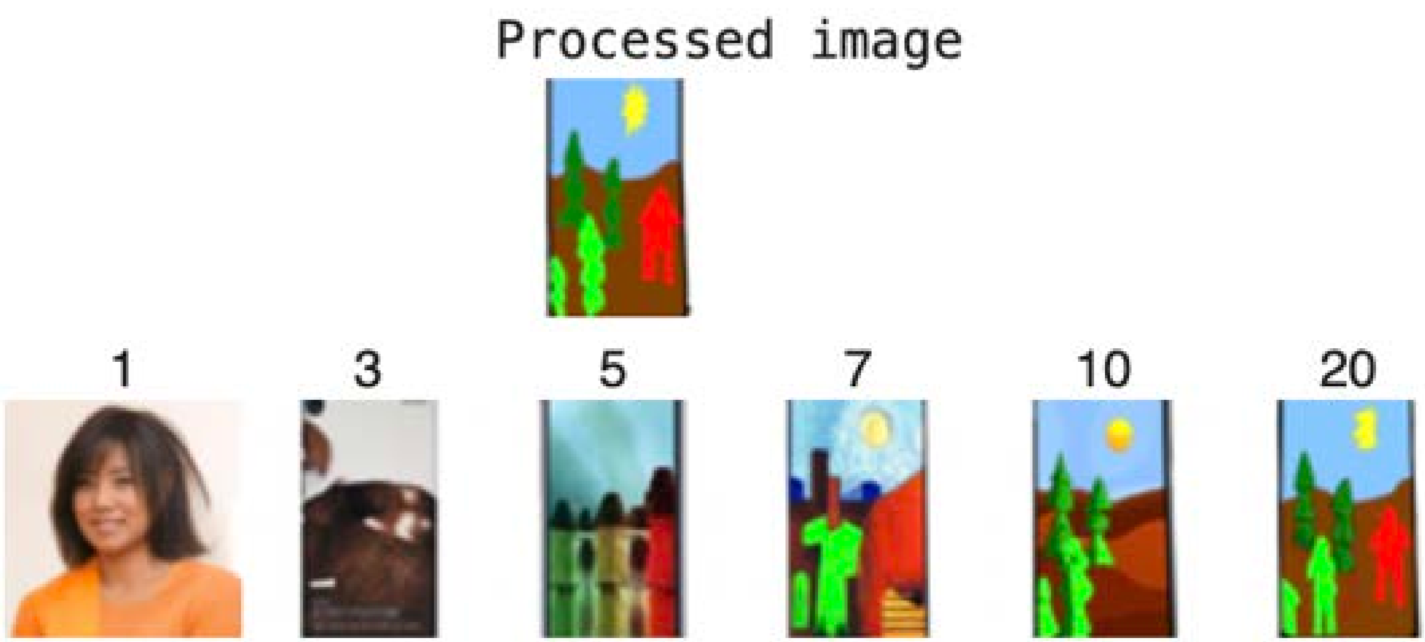
Part 1.6: Classifier-Free Guidance

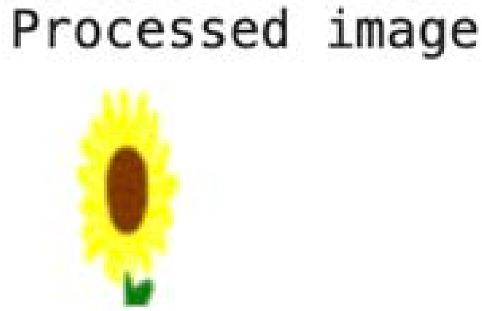
Below are 5 images generated with a `CFG` scale of 7:

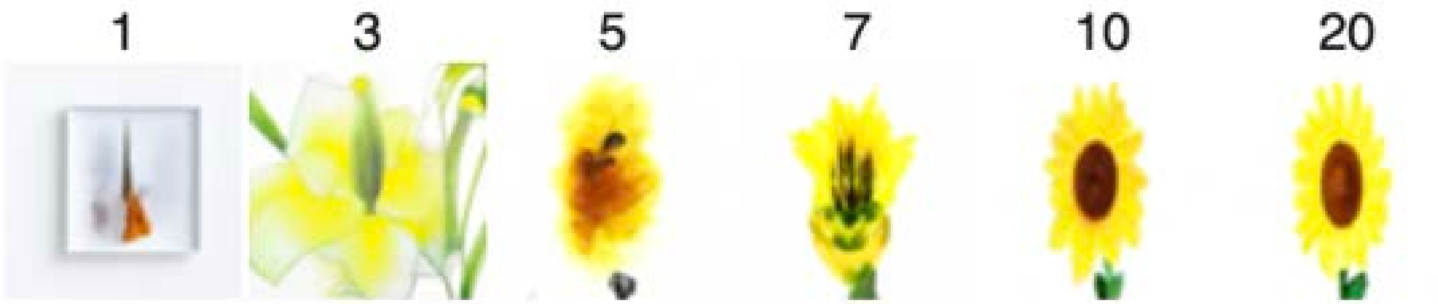
Part 1.7: Image-to-image Translation

Below are edits of the campanile, generated with noise levels `[1, 3, 5, 7, 10, 20]`:

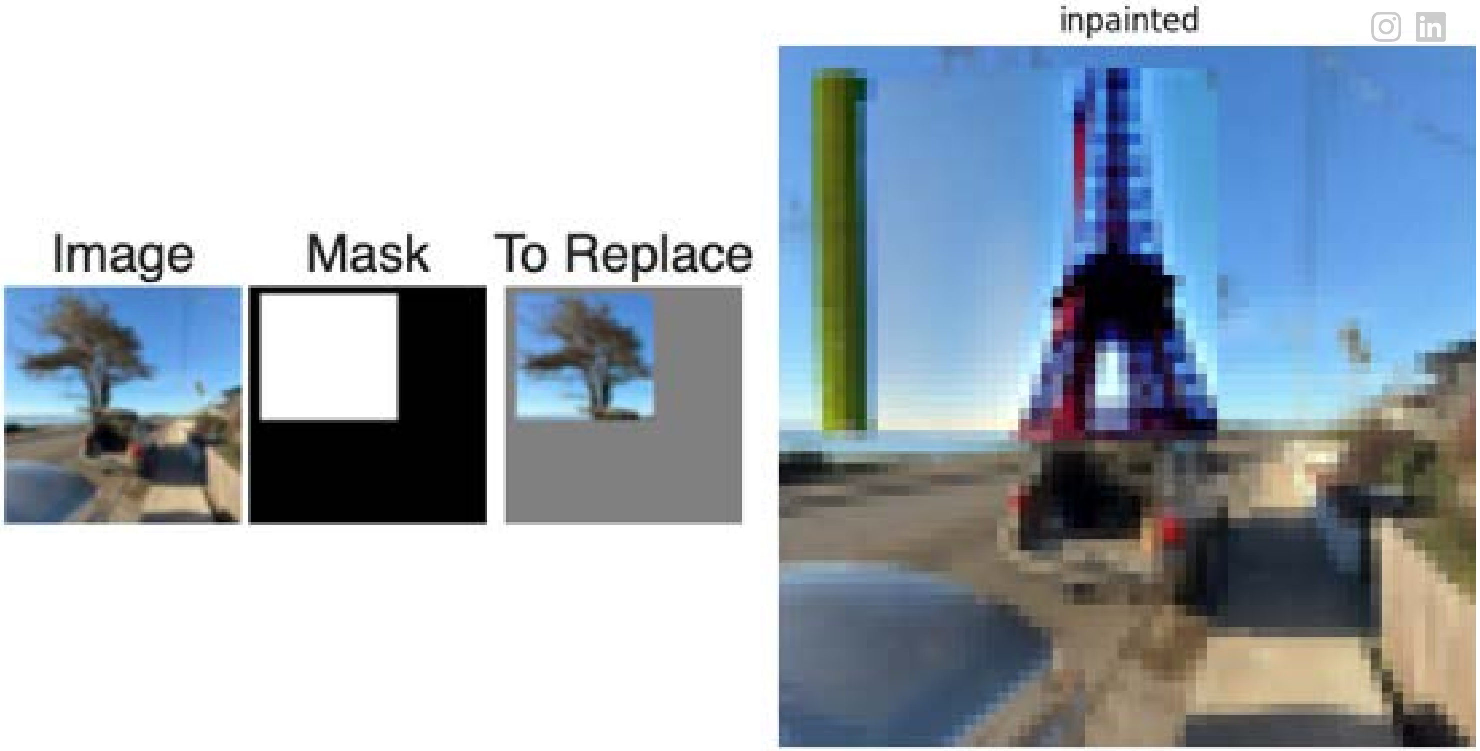
Below are three edits of hand-drawn and web images, using the same noise levels:

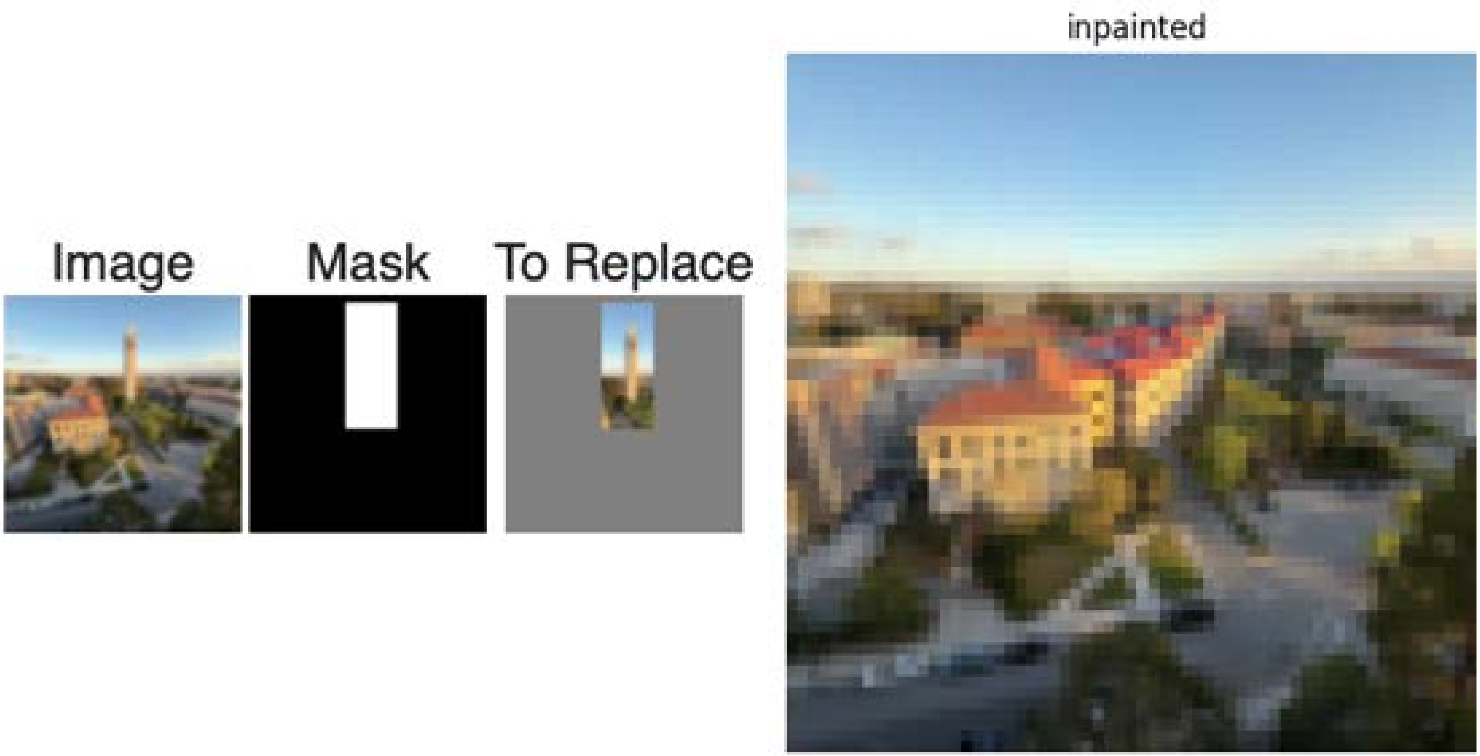




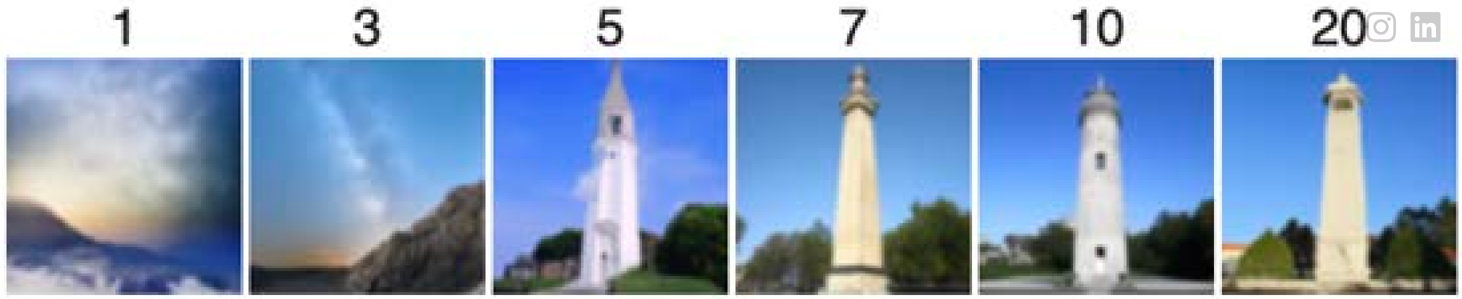


Below are three examples of inpainting:





Below are three edits that guide the projection with a text prompt:



"a photo of the night sky"

"an oil painting of redwood trees"

"a photo of mountains behind a lake"

Part 1.8: Visual Anagrams

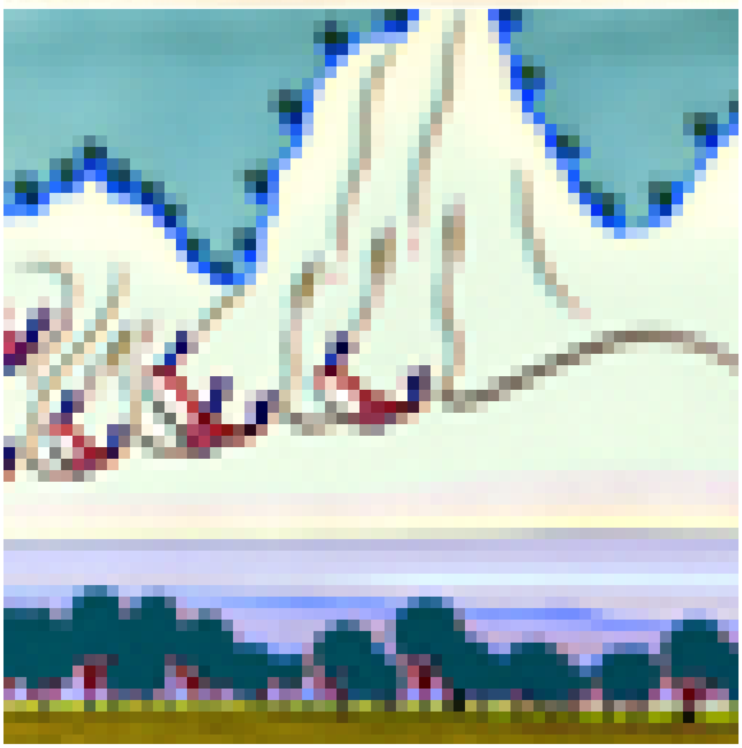
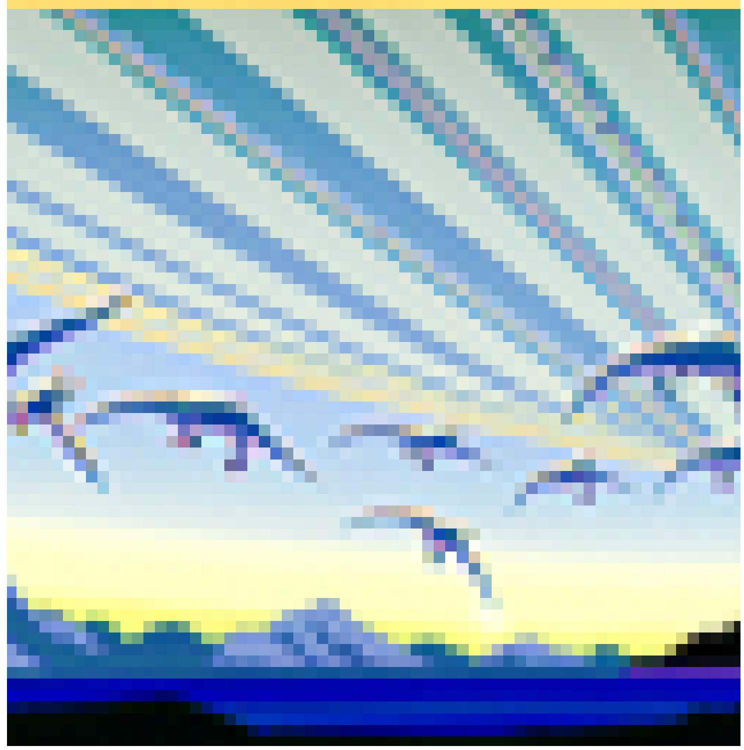
"a photo of mountains behind a lake" & "a photo of a person wearing sunglasses"

"an oil painting of a sailboat" & "a still life oil painting of flowers in a vase"

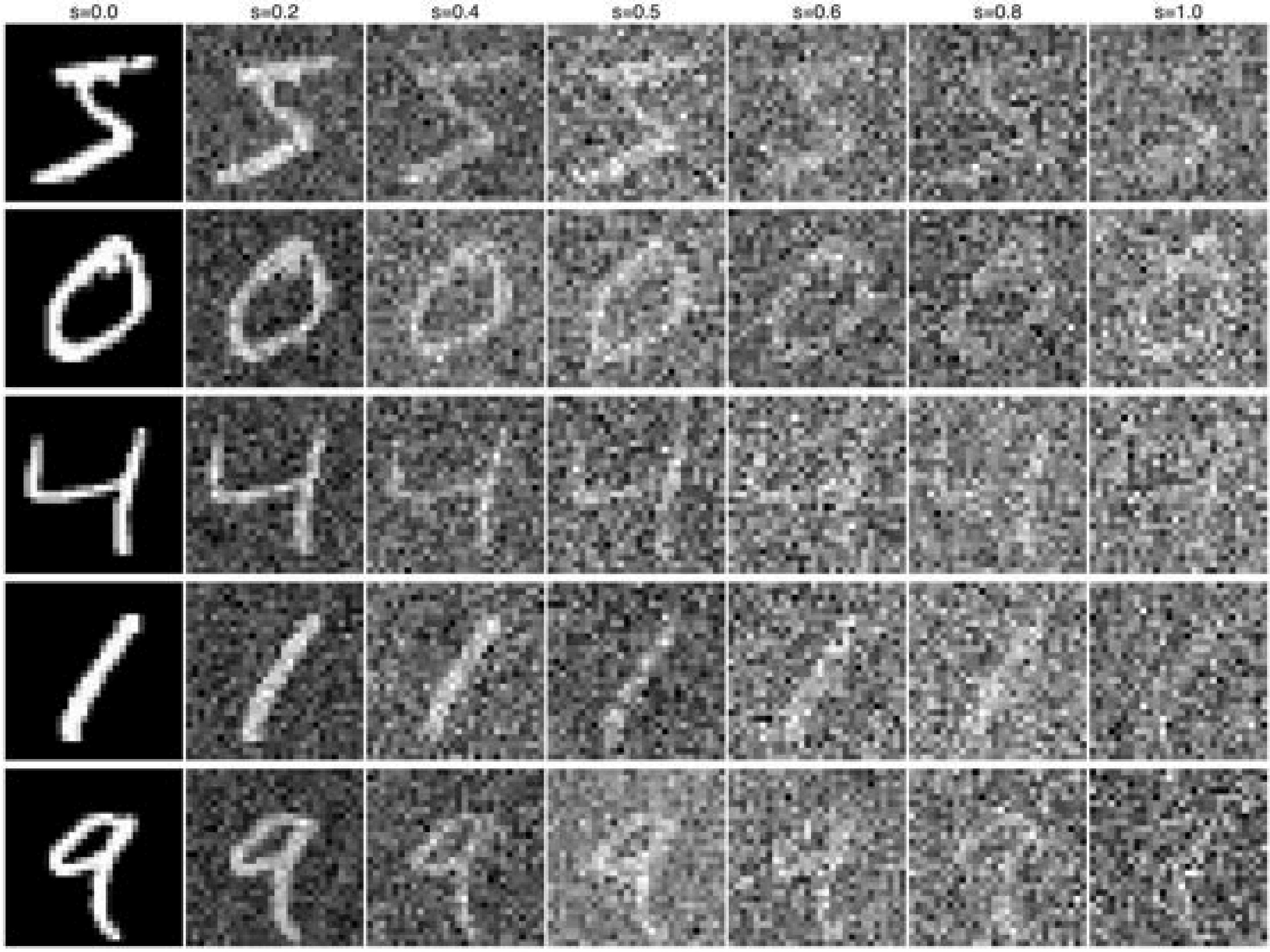
Part 1.9: Hybrid Images

Below are four hybrid images of `"a lithograph of clouds"` and `"a lithograph of a dinosaur"`:

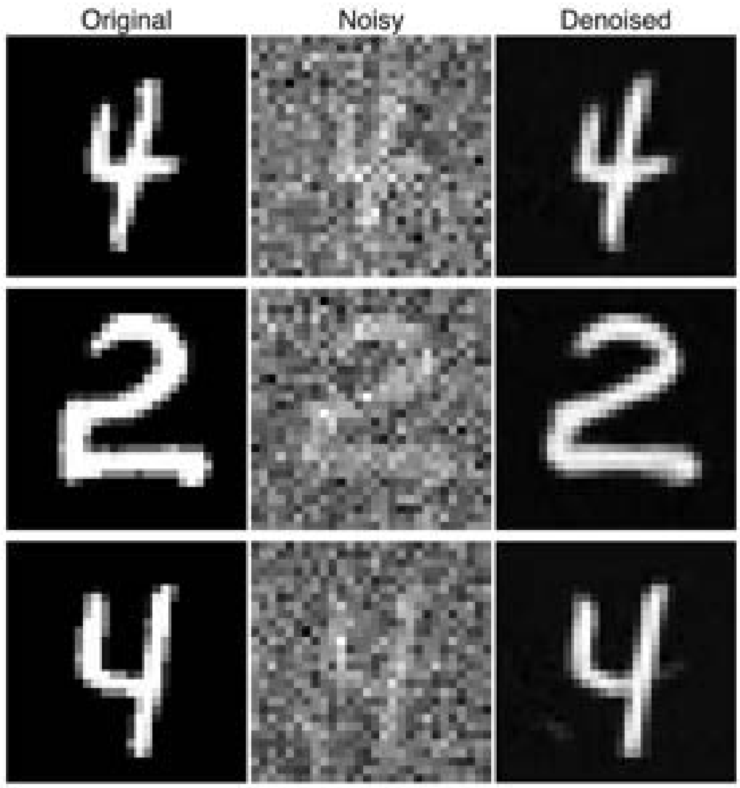


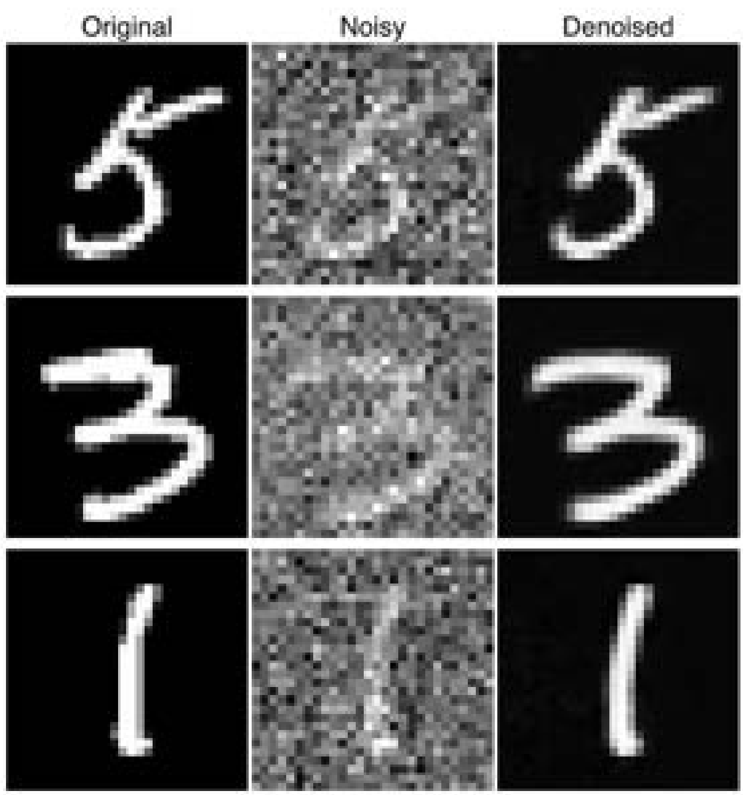
Part 2.2: Using the UNet to Train a Denoiser



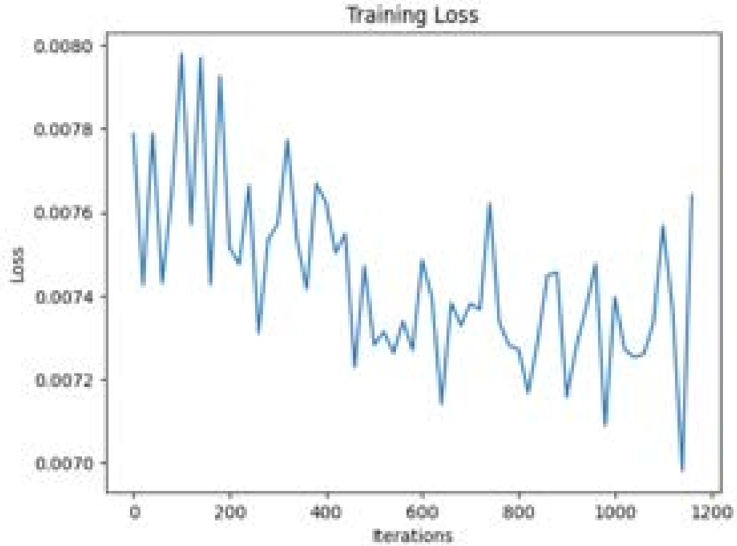
Part 2.2.1: Training



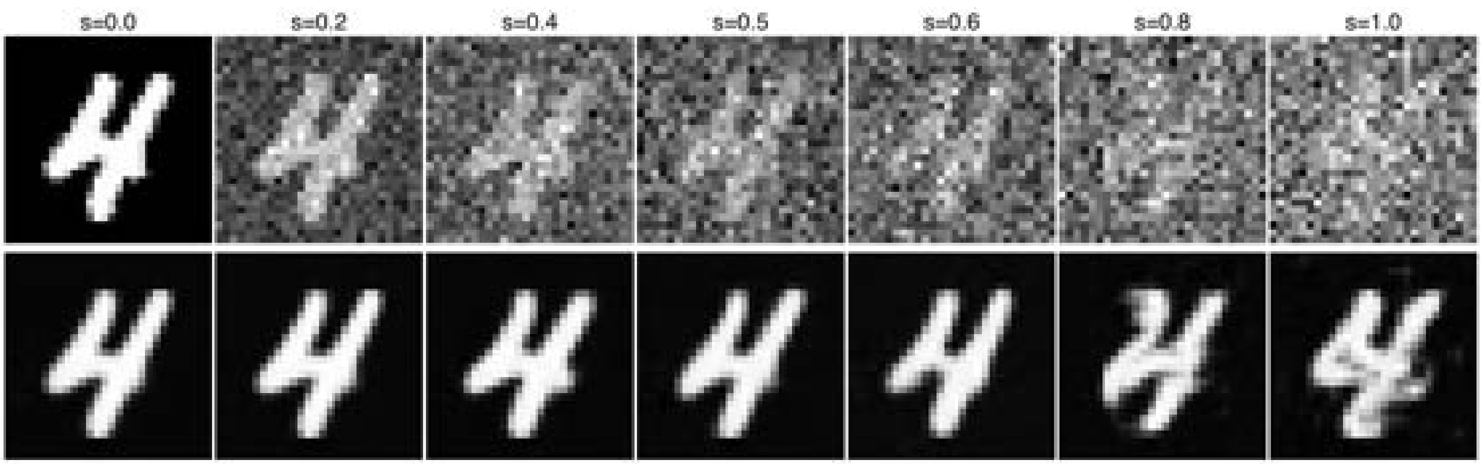
Sample results on the test set after the 1st epoch



Sample results on the test set after the 5th epoch

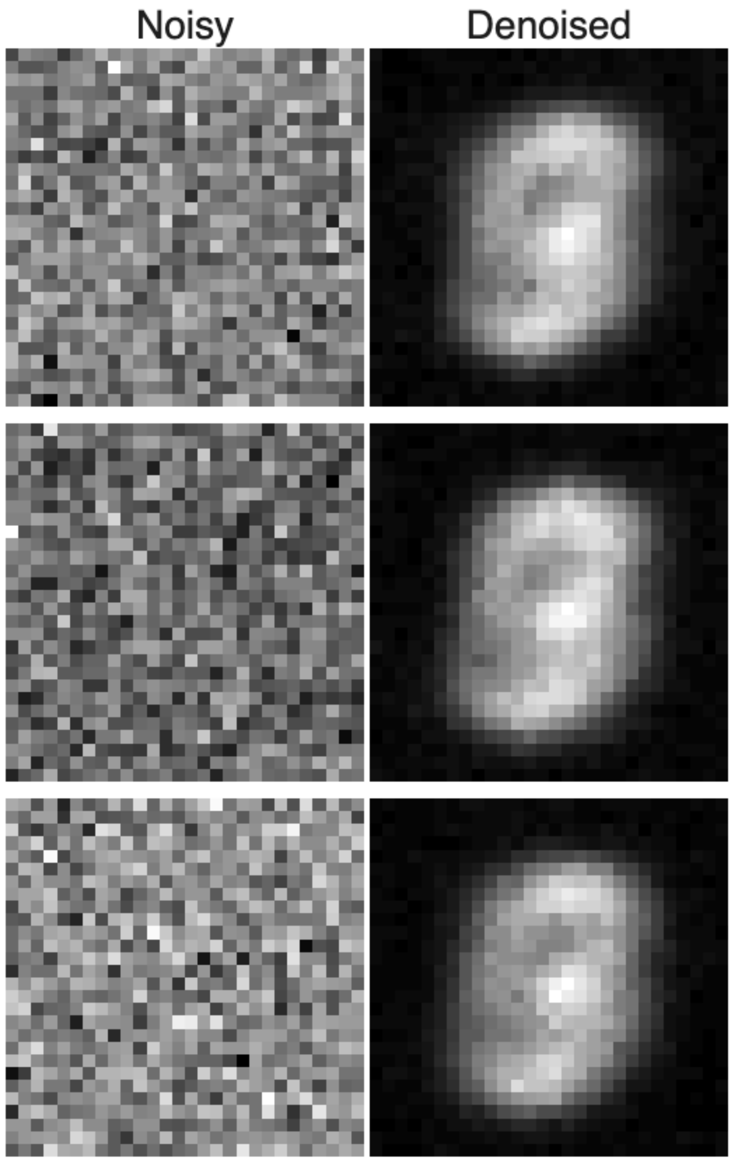


Part 2.2.2 Out-of-Distribution Testing

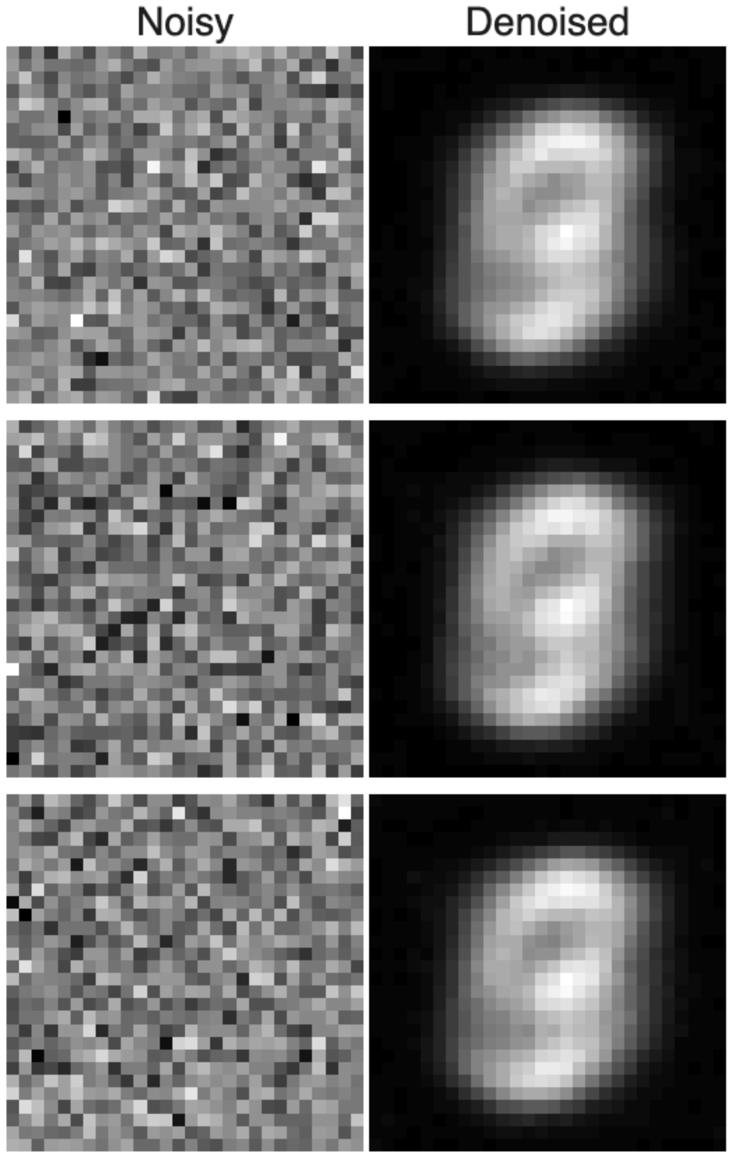


Part 2.2.3 Denoising Pure Noise

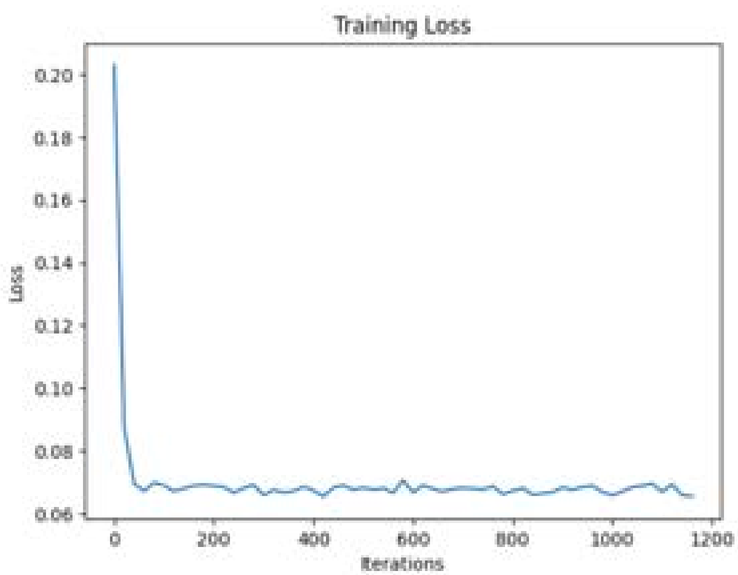
I observed that the generated outputs all followed the same format of an ambiguous number which followed the general shape of a 3, 5, 6, 8, 9, 0.



Sample results on the test set after the 1st epoch

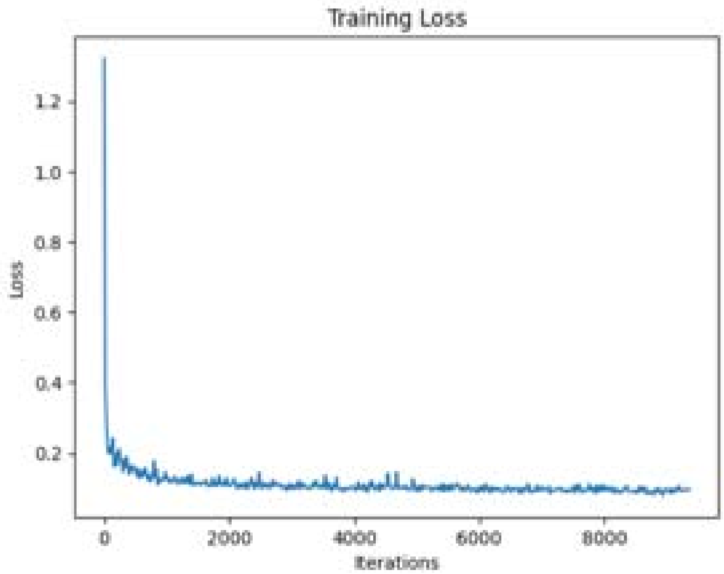


Sample results on the test set after the 5th epoch



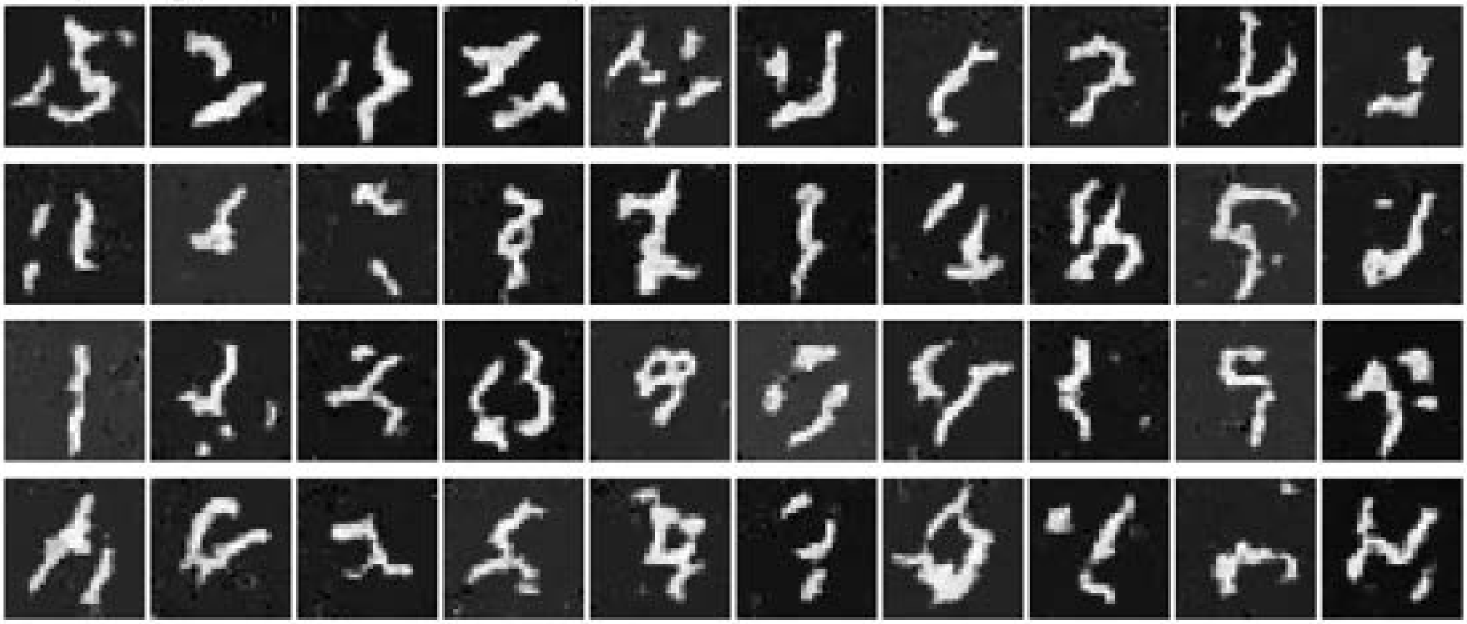
Part 3 (formerly Part 2): Training a Flow Matching Model

Part 3.2: Training the Time-Conditioned UNet

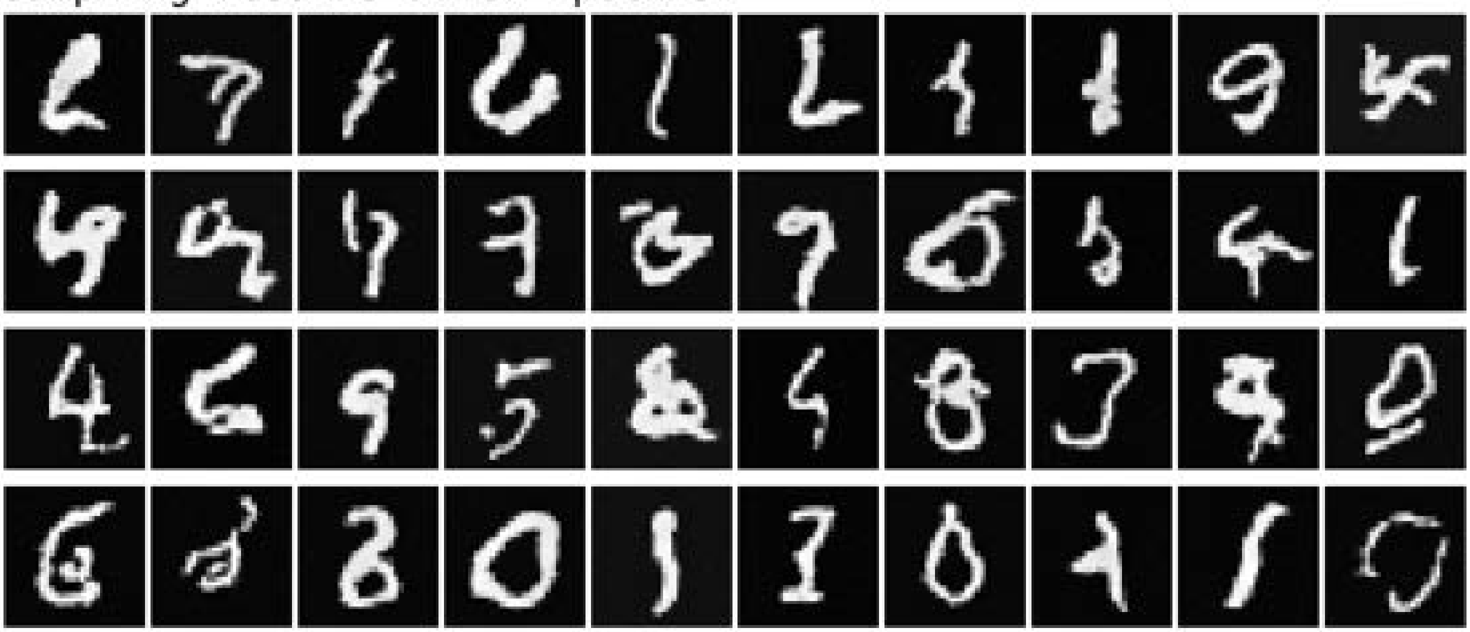


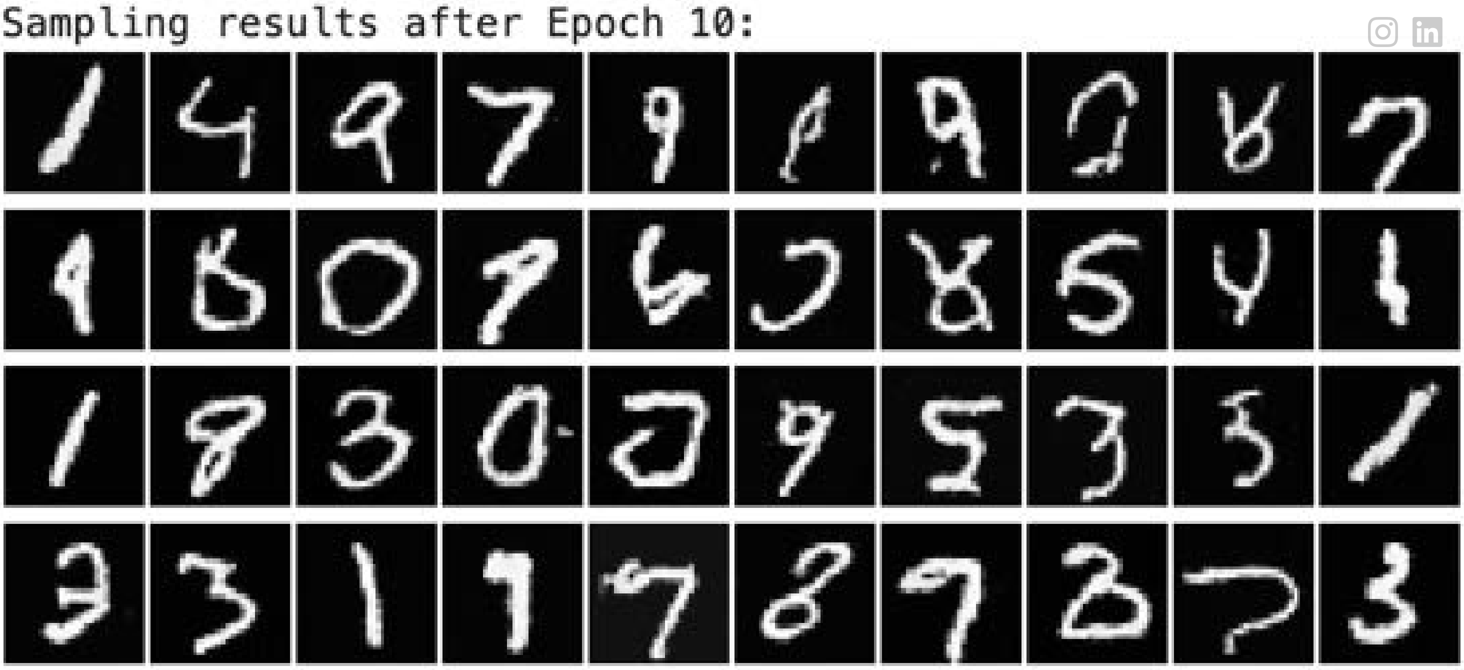
Part 3.3: Sampling from the UNet

Sampling results after Epoch 1:

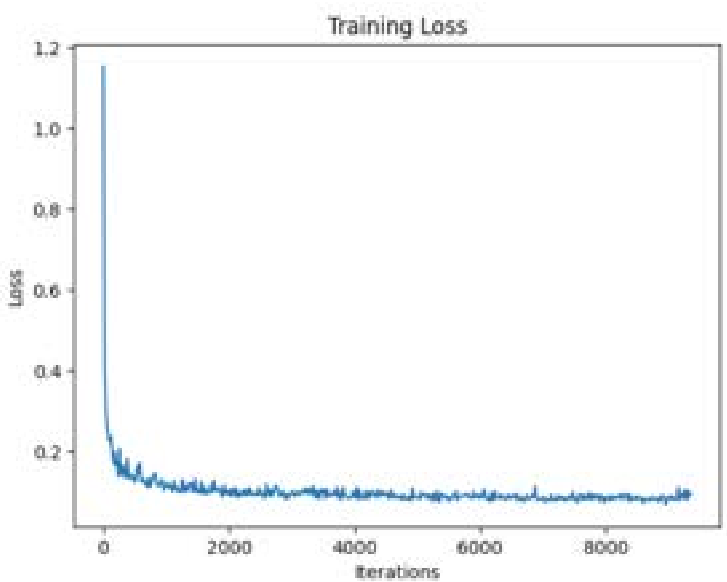


Sampling results after Epoch 5:

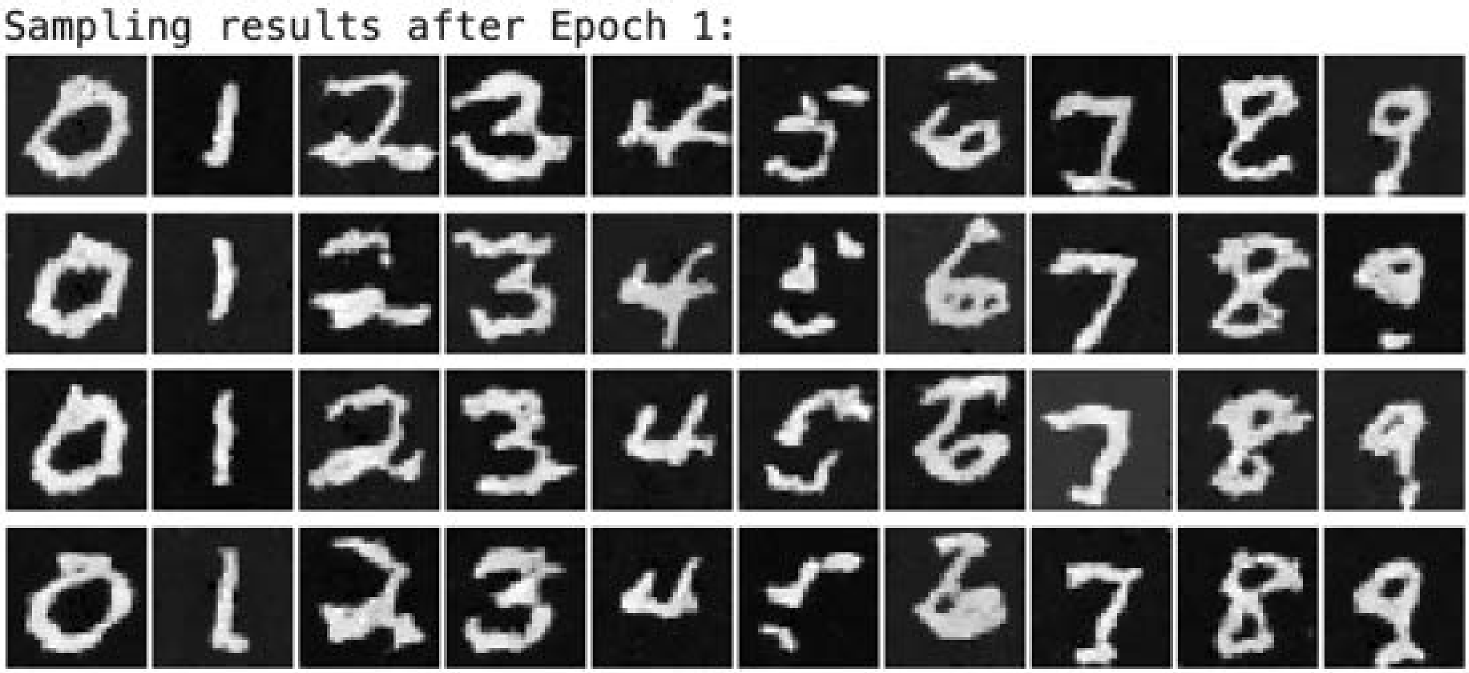


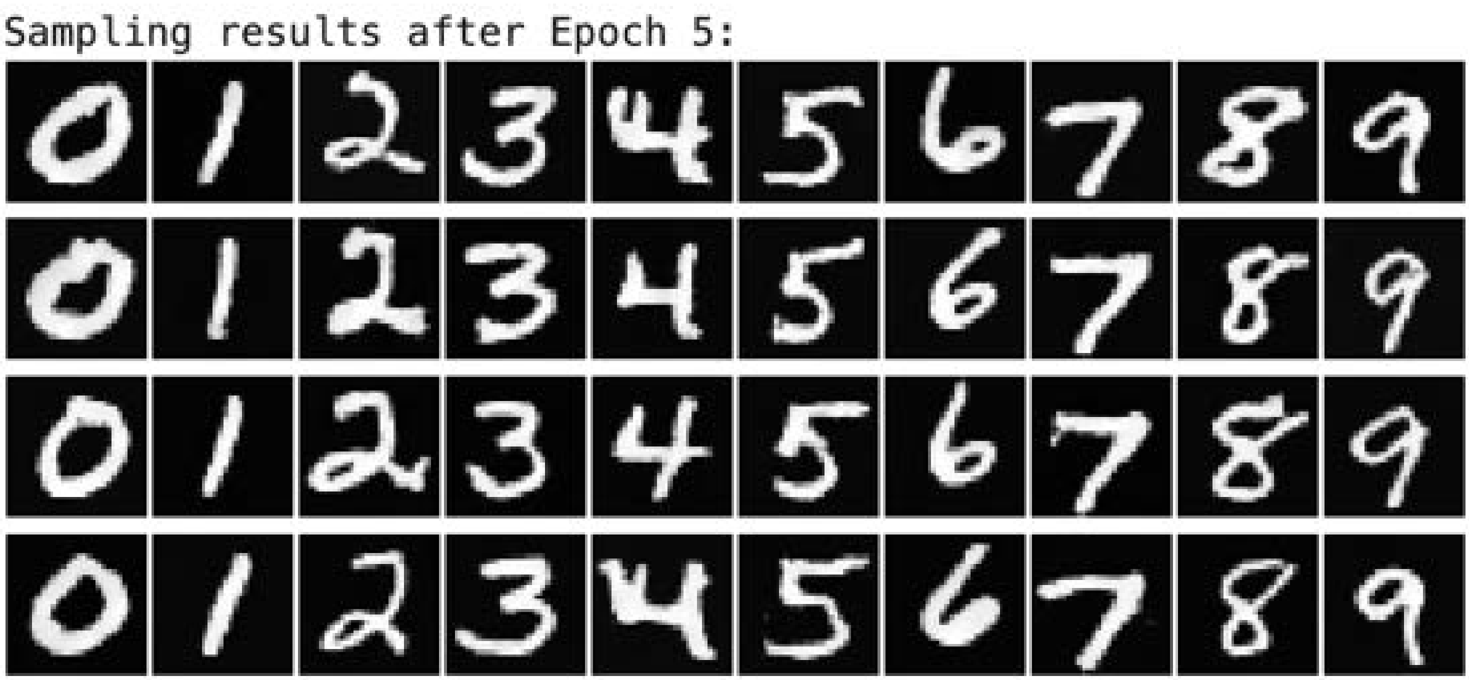


Part 3.5: Training the Class-Conditioned UNet



Part 3.6: Sampling from the UNet





Sampling results after Epoch 10: